UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 7: Applications of the First Moment Method

Content:

1. Random k-SAT
2. Longest Increasing Subsequences
3. Bond Percolation

7.1 Random k-SAT

Given n Boolean variables $x_1, \cdots, x_n \in \{0,1\}$, first sample k indices $\{i_1, \cdots, i_k\}$ uniformly at random. Then sample k i.i.d. Rademacher r.v. $\varepsilon_1, \cdots, \varepsilon_k$. Combining those, we get a k-clause:

$$e := (x_{i_1}\varepsilon_1) \vee \cdots \vee (x_{i_k}\varepsilon_k)$$

Here, if $\varepsilon_1 = 1$, then x_{i_1} appears as is in the clause e; otherwise it appears in negated way.

Logical OR: Recall the basic properties of logical OR: $u \vee v = 1$ iff either u or v is 1. Namely, unless u and v are both zeros, the logical OR is satisfied.

Now sample M ($M = \alpha n$) k-clauses $e_1, \cdots, e_M$ independently, and consider the formula: $\Phi := e_1 \wedge \cdots \wedge e_M$.

We are interested in the number of the assignments $x = (x_1, \ldots, x_n) \in \{0,1\}^n$ such that $\Phi(x) = 1$. Note that for $\Phi(x)$ to be 1, x must satisfy

Theorem 7.1.1 [Franco-Paull'83] If $\alpha > 2^k \ln 2$, then with high probability, there are no satisfying solutions as $n \to \infty$.

Proof: Denote

$$Z := |\{x \in \{0,1\}^n : \Phi(x) = 1\}| = \sum_{x \in \{0,1\}^n} \mathbf{1}\{\Phi(x) = 1\}$$

as the number of the solutions of $\Phi(x_1, \cdots, x_n) = 1$. To prove $\mathbb{P}(Z = 0) \to 1$, it suffices to show that $\mathbb{P}(Z > 0) \to 0$. By first moment method, we have

$$\mathbb{P}(Z > 0) = \mathbb{P}(Z \geq 1) \leq \mathbb{E}[Z]$$

then we just need to control $\mathbb{E}[Z]$:

$$\begin{aligned}
\mathbb{E}[Z] &= \sum_{x \in \{0,1\}^n} \mathbb{P}(\Phi(x) = 1) = 2^n \cdot \mathbb{P}(\Phi(x) = 1) \\
&= 2^n \mathbb{P}(x \text{ satisfies } e_1, \cdots, e_M) \\
&= 2^n [\mathbb{P}(x \text{ satisfies } e_1)]^M \\
&= 2^n [1 - \mathbb{P}(x \text{ doesn't satisfy } e_1)]^M \\
&= 2^n \left(1 - \frac{1}{2^k}\right)^M = 2^n \left[\left(1 - \frac{1}{2^k}\right)^{2^k}\right]^{M/2^k}, \\
&\leq 2^n \exp\left(-\frac{M}{2^k}\right) = \exp\left[n \ln 2 - \frac{\alpha n}{2^k}\right] \\
&= \exp\left[-\frac{n}{2^k}(\alpha - 2^k \ln 2)\right] \to 0
\end{aligned}$$

using the fact $\alpha > 2^k \ln 2$. Above, we used the fact that a fixed $x \in \{0,1\}^n$ satisfies a randomly generated clause with probability $1 - 2^{-k}$ (basically, the only way a clause is not satisfied is if all literals present are zeros).

7.2 Longest Increasing Subsequences

Let $\sigma = (\sigma(1), \cdots, \sigma(n))$ be a random permutation of $\{1, \cdots, n\}$. A subsequence $i_1 < \cdots < i_k$ is called increasing if $\sigma(i_1) < \cdots < \sigma(i_k)$. Denote $L_n = L_n(\sigma)$ as the length of the longest increasing subsequence of σ. When σ is chosen uniformly at random (among all $n!$ choices), we have

Theorem 7.2.1 $\mathbb{E}[L_n] \le c\sqrt{n}$.

Proof: For any l s.t. $0 \le l \le n$, we have

$$L_n = L_n \mathbf{1}\{L_n < l\} + L_n \mathbf{1}\{L_n \ge l\} \le l\mathbf{1}\{L_n < l\} + n\mathbf{1}\{L_n \ge l\}$$

$$\Rightarrow \ \mathbb{E}[L_n] \le l\mathbb{P}(L_n < l) + n\mathbb{P}(L_n \ge l) \le l + n\mathbb{P}(L_n \ge l)$$

To control $\mathbb{P}(L_n \ge l)$, we still use first moment method. Denote $Z_l = Z_l(\sigma)$ as

$$Z_l := |\{(i_1, \cdots, i_l) : i_1 < \cdots < i_k, \sigma(i_1) < \cdots < \sigma(i_k)\}|$$

$$= \sum_{1 \le i_1 < \cdots < i_k \le n} \mathbf{1}\{\sigma(i_1) < \cdots < \sigma(i_k)\}$$

then we have

$$\mathbb{P}(L_n \ge l) = \mathbb{P}(Z_l \ge 1) \le \mathbb{E}[Z_l] = \sum_{1 \le i_1 < \cdots < i_k \le n} \mathbb{E}[\mathbf{1}\{\sigma(i_1) < \cdots < \sigma(i_k)\}]$$

$$= \binom{n}{l} \mathbb{P}(\sigma(i_1) < \cdots < \sigma(i_k)) = \binom{n}{l} \cdot \frac{1}{l!}$$

$$= \frac{n(n-1)\cdots(n-l+1)}{(l!)^2} \le \frac{n^l}{(l!)^2}$$

by Stirling's formula, we know $l! \ge \sqrt{2\pi l}(l/e)^l$, then

$$\mathbb{E}[L_n] \le l + n\mathbb{P}(L_n \ge l) \le l + n\mathbb{E}[Z_l] \le l + n \cdot \frac{n^l}{(l!)^2}$$

$$\le l + \frac{n^{l+1}}{2\pi l} \left(\frac{e}{l}\right)^{2l} = l + \frac{n}{2\pi l} \left(\frac{e\sqrt{n}}{l}\right)^{2l}$$

Let $l = c\sqrt{n}$ with $c > e$, the upper bound becomes

$$c\sqrt{n} + \frac{\sqrt{n}}{2\pi c} \left(\frac{e}{c}\right)^{2c\sqrt{n}} = c\sqrt{n} + e^{-\theta(\sqrt{n})}$$

so $\mathbb{E}[L_n] \le c\sqrt{n} + e^{-\Theta(\sqrt{n})}$ for any $c > e$.

Remark. *In fact, it is known that $\mathbb{E}[L_n] = \Theta(\sqrt{n})$. The lower bound is a simple combinatorial argument, originally due to Erdos and Szekeres. The hardest part*

of this model was to pin down the correct constant in front of $\sqrt{n}$. It is known that

$$\mathbb{E}[L_n] = 2\sqrt{n} + cn^{1/6} + o(n^{1/6})$$

as $n \to \infty$, where $c \approx -1.77$. This result is of course beyond the scope of our course.

7.3 Bond Percolation

Denote $\mathbb{L}^2$ as the lattice points in $\mathbb{Z}^2$, consider the graph $G = (\mathbb{L}^2, E)$, where $E = \{(u, v) \in \mathbb{Z}^2 \times \mathbb{Z}^2 : \|u - v\|_1 = 1\}$ are the edges between the lattice points. Suppose that each edge is open with probability p independently. Pour some water at the origin. Would it flow to infinity?

When $p = 0$, all edges are blocked, and there is no flow. When $p = 1$, all edges are open, and the water can flow to infinity. Given $x \in \mathbb{Z}^2$, we write $y \sim x$ if y is reachable from x through open edges, and denote $C_x = \{y \in \mathbb{Z}^2 : y \sim x\}$ as the reachable points from x. Define

$$\theta(p) = \mathbb{P}_p(|C_o| = +\infty)$$

as the probability of the event that the origin has infinite reachable points when each edge is open with probability p. Define

$$P_c(\mathbb{L}^2) := \sup\{p \geq 0 : \theta(p) = 0\}$$

as the critical probability of the phase transition.

Theorem 7.3.1 If $p_1 \leq p_2$, then $\theta(p_1) \leq \theta(p_2)$.

Proof: For any $e \in E$, let $U_e \sim \text{Unif}[0, 1]$. In the first model, keep edge e open if $U_e \leq p_1$; in the second model, keep edge e open if $U_e \leq p_2$. every edge open in the first model is also open in the second model. Thus, C_o in the first model is a subset of C_o in the second model, so $\theta(p_1) \leq \theta(p_2)$.

Exercise. Show that $P_c(\mathbb{L}^1) = 1$.

Theorem 7.3.2 $P_c(\mathbb{L}^2) \in (0, 1)$.

Proof: We first show that $P_c \geq 1/3$. Denote Z_n as the number of the self-

avoiding open paths starting from the origin with length n. Fix $p < 1/3$:

$$\begin{aligned}\mathbb{P}_p(|C_o| = +\infty) = \mathbb{P}_p\left(\bigcap_{n\geq 1}\{Z_n > 0\}\right) &= \lim_{n\to\infty}\mathbb{P}_p(Z_n > 0) \\ &\leq \lim_{n\to\infty}\mathbb{E}[Z_n] \leq \lim_{n\to\infty}\left(4\cdot 3^{n-1}\cdot p^n\right) = 0\end{aligned}$$

If $p < P_c$, then by theorem 7.3.1, $\theta(p) = \mathbb{P}_p(|C_o| = +\infty) = 0$, and

$$\mathbb{P}_p\left(\exists x \in \mathbb{Z}^2 : |C_x| = +\infty\right) \leq \sum_{x\in\mathbb{Z}^2}\mathbb{P}_p(|C_x| = +\infty) = 0$$

If $p > P_c$, then $\mathbb{P}_p(|C_o| = +\infty) > 0$, and $\mathbb{P}_p(\exists x \in \mathbb{Z}^2 : |C_x| = +\infty) = 1$ by Kolmogorov 0-1 law.

(**To be continued in next lecture.**)